FRC | Science Fair | Vacuum Robot

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Brief Overview of Events

FRC Magnolia Regional (Mississippi)
3/18 - 3/21



Started using API keys to grab standard information about teams, as well as more information and input sections

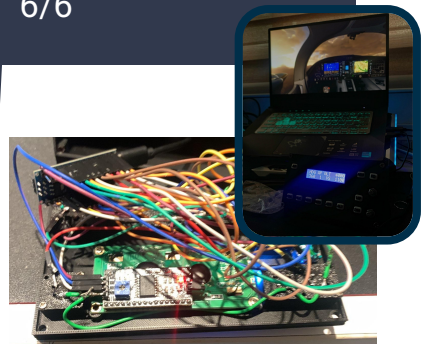
Created the first version of my Pit Collect website and put it to use.



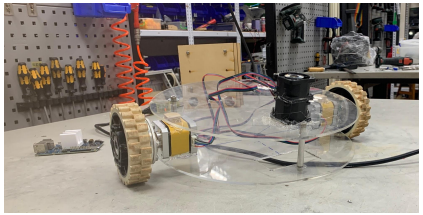
FRC Championship (Houston)
4/29 - 5/1

Built a mini autopilot to control the autopilot in my flight simulator.

Mini Autopilot Panel
6/6



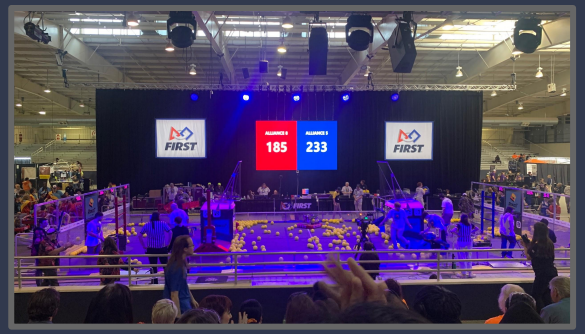
Built a mini autopilot to control the autopilot in my flight simulator.



Vacuum Robot & Science Fair
7/1 - Present


803

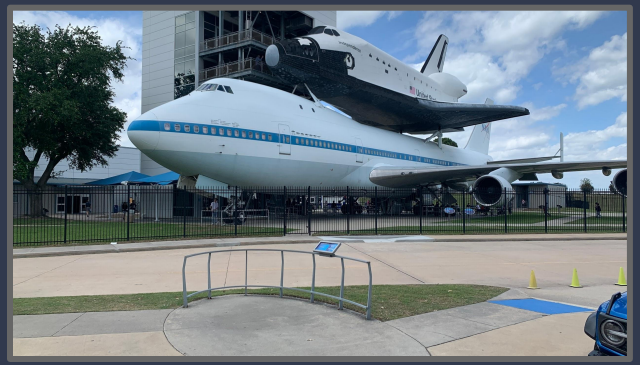

GR
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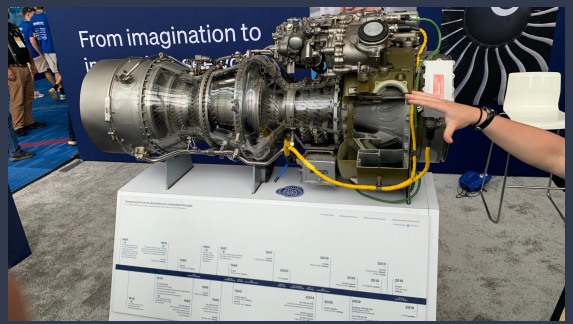

FIRST
ALLIANCE 1
185
ALLIANCE 2
233
FIRST




6998

FRC




From imagination to


FIRST 2026 CHAMPIO

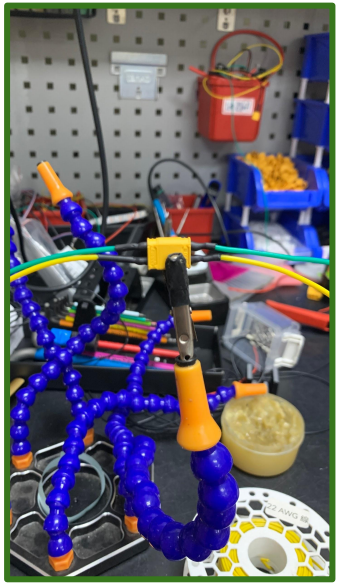
2026 Robot

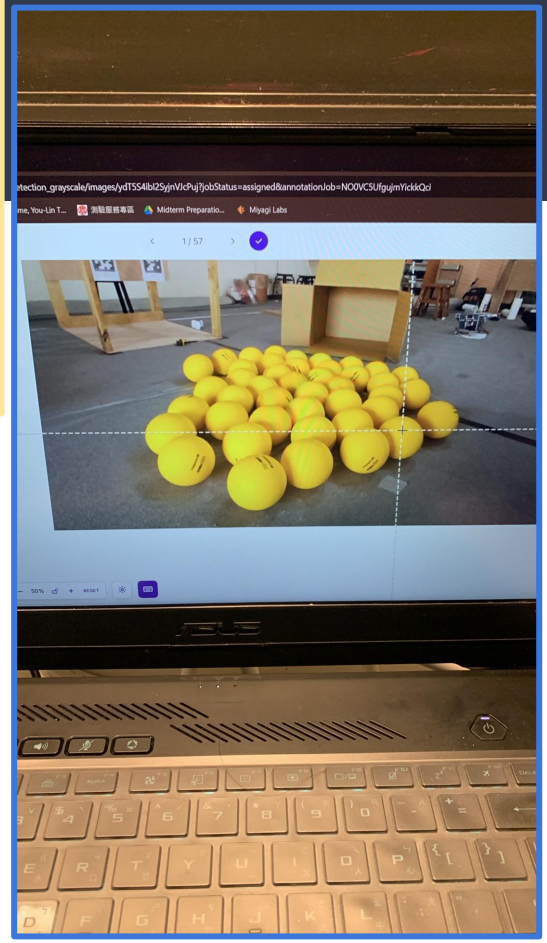


Contributions

- Created a type of scouting website called Pit Collect, which our team used to analyze the hardware and basic strategy of other teams
  - Can be used to figure out how to play against them
  - Determine how to cooperate with them when they are on the same side as us
  - Helped to build an advanced knowledge and understanding of HTML, JS, and CSS through the help of vibe coding and my own code.
- Helped solder part of the wiring for the robot
  - Massively developed my soldering skills
  - Paved the way for future projects
- Worked on object detection models for the 2026 robot
  - Helped to map out game pieces and compile into an object detection model for our robot
  - Taught me the basics of object detection and object detection software, such as Roboflow







FRC Magnolia Regional

3/18 - 3/21



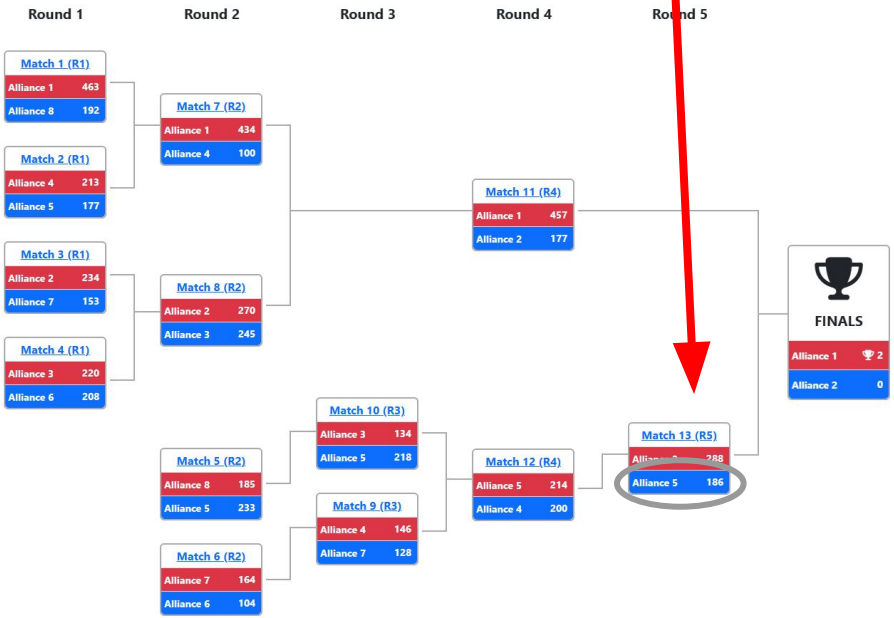
Results



Final Score:
188:288

- 12th out of 46 teams
- Match record of 6-3-0.
- Got into the semifinals of the playoff matches
- Won the Impact Award
  - took us to the FRC Championship in Houston
- My Pit Collect Website was used to track the data of over 40 teams.

Rank	Team	Ranking Score	Avg Match	Avg Auto Fuel	Avg Tower	Record (W-L-T)	DQ	Played	Total Ranking Points*
1	4400	4.11	264.11	40.11	0.00	9-0-0	0	9	37
2	8044	4.00	252.56	53.56	1.11	9-0-0	0	9	36
3	4087	3.33	211.11	44.11	0.00	7-2-0	0	9	30
4	364	3.22	166.89	34.11	0.00	7-2-0	0	9	29
5	2183	3.22	148.78	35.89	0.00	7-2-0	0	9	29
6	2556	3.00	160.56	29.56	0.00	6-3-0	0	9	27
7	3753	3.00	141.78	25.33	0.00	7-2-0	0	9	27
8	2992	2.89	164.78	29.11	1.11	6-3-0	0	9	26
9	3039	2.78	160.56	33.11	0.00	6-3-0	0	9	25
10	1912	2.78	145.22	29.22	0.00	7-2-0	0	9	25
11	6184	2.67	156.33	21.22	0.00	6-3-0	0	9	24
12	6998	2.67	152.56	33.78	0.00	6-3-0	0	9	24
13	9309	2.67	102.00	22.33	0.00	7-2-0	0	9	24
14	10661	2.56	170.00	19.33	0.00	5-4-0	0	9	23
15	4336	2.11	94.33	21.89	0.00	5-4-0	0	9	19
16	456	2.00	124.56	21.33	0.00	4-5-0	0	9	18
17	9503	2.00	108.44	25.56	0.00	4-5-0	0	9	18
18	10101	2.00	81.33	19.33	1.11	5-4-0	0	9	18



Struggles/Problems

QR Code too large

- To transfer data from the device the scouting was done on to the device used to compile all the results, we decided to use a qr code to transfer all the values

- In the early stages of development, the QR code was too large, resulting in the scanner not being able to pick up the QR code

Solution:

Burned Wires

- When I was soldering wires, I sometimes used to accidentally burn the plastic/insulation of the wire and wire connectors, rendering the wires useless.
- The soldering tin would not melt properly, and so I would keep the soldering iron on the component for a long time.

Solution:

QR Code Fix

To solve the issue of my QR code being too large, I utilized 4 main methods:

1. Setting Visual Dimension Limits
   a. Set the QR code size to 256 by 256 px
      i. Ensures the image size will not change
2. LZ-string Compression
   a. Before any of the qr codes are generated, all the TSV(tab separated values) are compressed into Base64 format
      i. E.g. A -> 000000, B -> 000001…
3. Character Sanitization
   a. Newlines and tabs are deleted from text inputs to save space

```
new QRCode(qrEl, {
    text: compressed, width: 256, height: 256,
    colorDark: "#000000", colorLight: "#ffffff",
    correctLevel: QRCode.CorrectLevel.L
});
```

```
const compressed = LZString.compressToBase64(tsvValues.join('\t'));
```

Doesn't this make the file size larger?
It does, but LZ string compression looks for recurring patterns and creates a dictionary for where the values should go

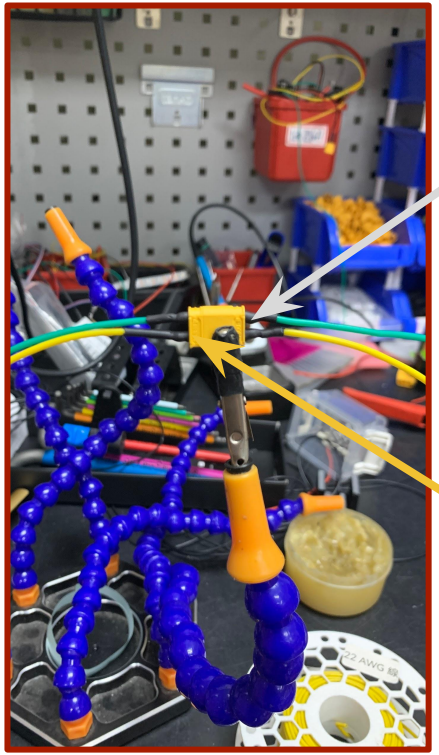
Back:

```
document.getElementById('notes').value.replace(/[\t\n]/g, ' ')
```

Soldering Skills Improvement

Since I kept the soldering iron on the component for too long, it caused the plastic surrounding the metal contacts to melt. This results in the component becoming unusable, which wastes time and resources.

1. Plug a spare connector in
    a. By plugging in a spare connector, it acts as a heat sink. The extra metal contacts mean that more heat is required to raise the temperature of the component, allowing me to keep the soldering iron on for longer.
2. Pre-tin both the connector and the wire
    a. I used to only pre-tin the wire, which meant that there will be poor heat transfer
    b. By tinning the connector too, the metal heats up way quicker



Back:

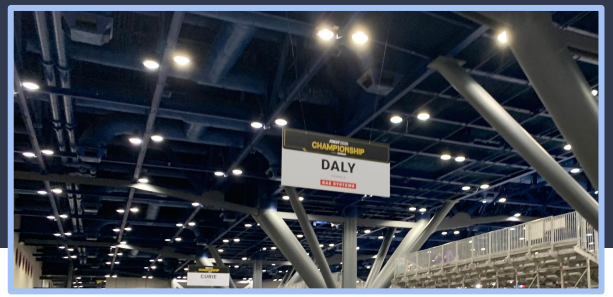
FRC Houston Championships

4/29 - 5/1



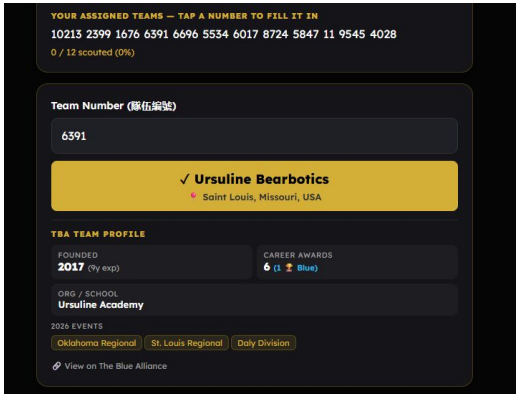


Results



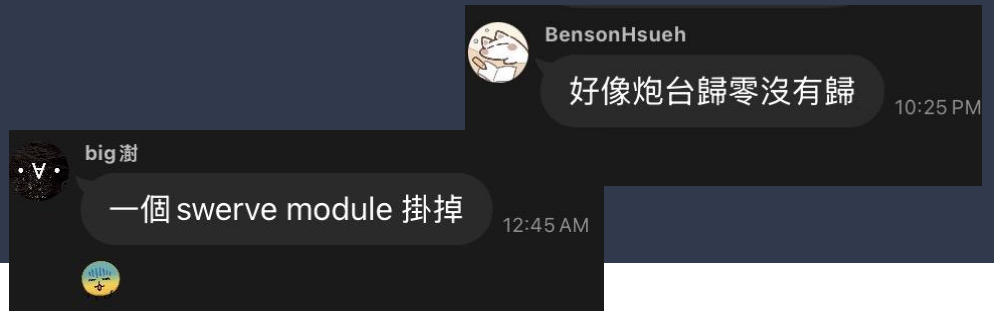
45		The Pascack PI-oneers	1.80	381.40	80.30	2.50	4 - 6 - 0	10
46	624	CRyptonite	1.80	361.60	74.50	4.00	4 - 6 - 0	10
47	4189	Chargers	1.80	360.90	74.10	4.50	4 - 6 - 0	10
48	2718	Team OKC e'possums (Σ 1/n! x 10^9)	1.80	351.00	73.80	1.00	4 - 6 - 0	10
49	6998	Unipards	1.80	340.80	62.20	0.00	4 - 6 - 0	10
50	5550	BRONChO BOTS	1.80	296.90	61.40	0.00	4 - 6 - 0	10
51	3880	Tiki Techs	1.70	290.80	60.40	1.00	5 - 5 - 0	10
52	1156	Under Control	1.60	387.90	73.80	1.50	3 - 6 - 0	10
53	1629	Garrett Coalition (GaCo)	1.60	350.90	69.10	2.50	4 - 6 - 0	10

- 9 divisions in total
- 49th out of 75 teams in the Daly Division
- My Pit collect website was used to scout over 70 international teams
- Win loss record of 4-6-0
- Added major features to my Pit Collect website
  - API key grabbing from the Blue Alliance to display basic information about a team.
  - Allowed for teams to draw out multiple autonomous strategies





Struggles/Problems

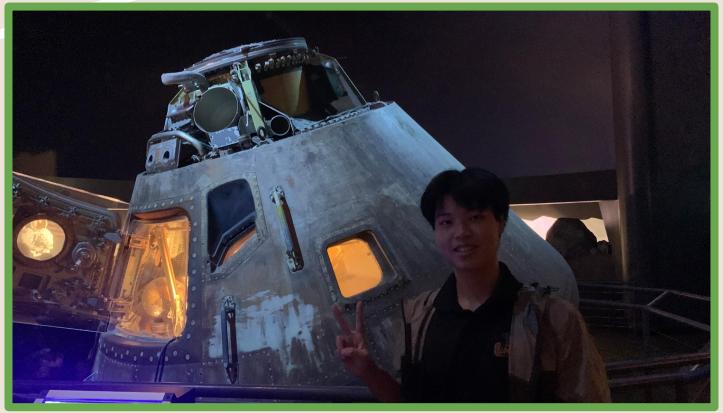


1. Robot wiring failures during match
    a. During a lot of the matches, our robot could not move due to electrical wiring mistakes and zeroing issues due to human error
    b. Time constraints
2. Autonomous path drawing is laggy
    a. To lower the QR code size, I decided to lower the amount of points(x,y) that will be plotted when you drew on the field.
    b. This made the line drawing inaccurate and laggy, which meant we were compromising our data for file size
    c. = lossy compression

1. Although I was not in the fixing team, this experience has taught me that even under pressure, you cannot skip steps in the standard procedure, and must follow the checklist as intended.
2. To fix the

Johnson Space Center











Saturn V Rocket

When I visited the Johnson Space Center in Houston, the thing that stood out to me the most was the Saturn V Rocket. When I approached it, I was in awe of its massive size and complex engineering.

Realization: True engineering requires everything to be done step by step, and everything has to be done to perfection.

Space Shuttle Carrier

- This modified Boeing 747 carrying the shuttle orbiter shows structural load distribution, as well as the aerodynamic modifications required for a B747 to transport the Space Shuttle Orbiter.
- I examined the old flight controls and avionics, where I noticed that it still resembles modern flight displays today.





Johnson Space Center Reflections

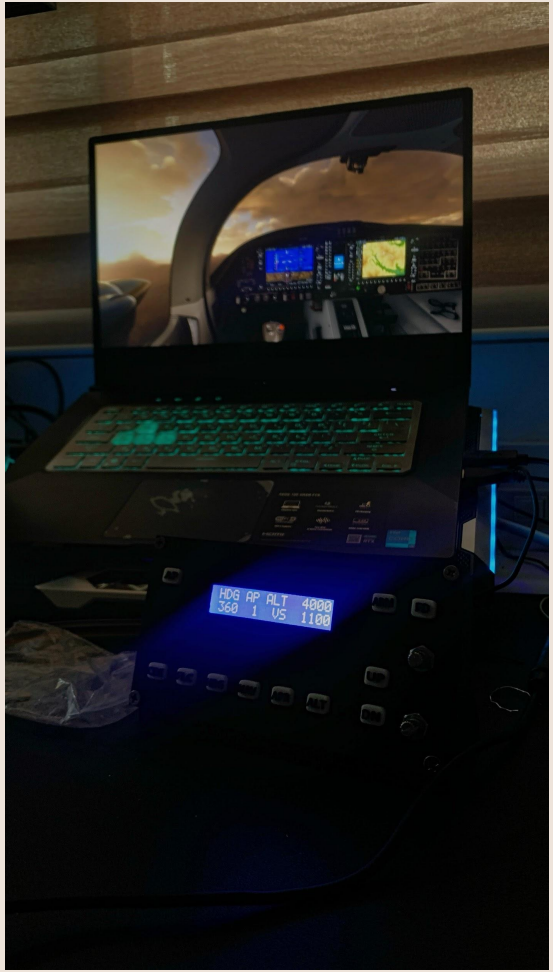
Seeing how mechanical, electrical, and control systems come together in Aerospace inspired me to build my own physical Flight Simulator Autopilot Panel. The large aircrafts and spacecrafts further inspired me to pursue the field of STEAM based majors.

Inspiration for Future Career

Further reinforced my goal to pursue EE/Mechanical Engineering/ Communications to become a pilot with a deep technical foundation in flight control systems

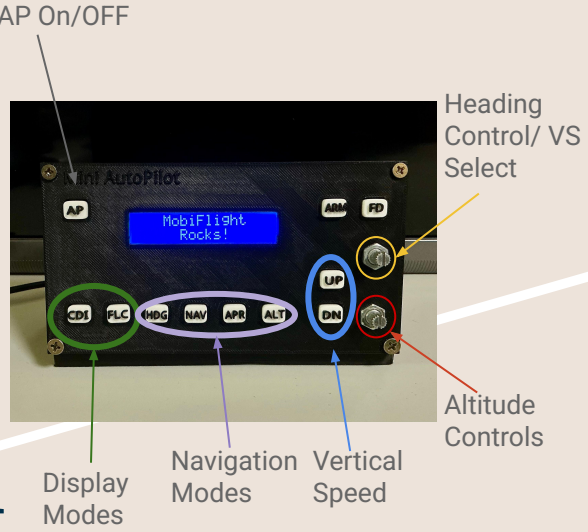


Flight Simulator Autopilot



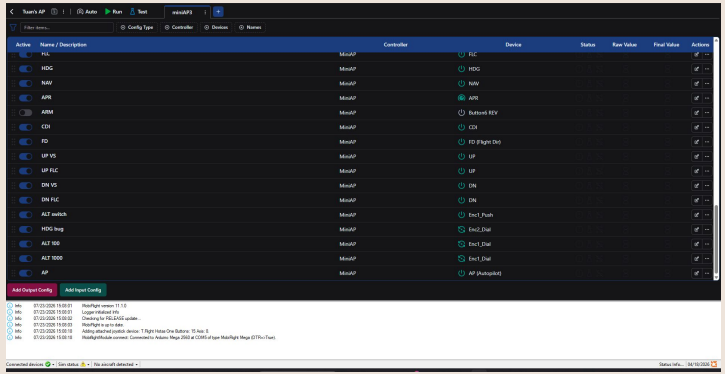
Overview & Functionality

- Physical autopilot hardware built to control heading, vertical speed, and navigation modes in flight simulation.
- Integrates real-time rotary encoder inputs and LCD outputs to display live in-game flight data.

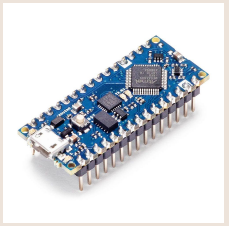


Software and Communication

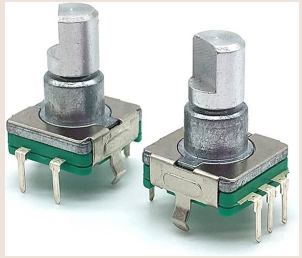
- Using Mobiflight, I created an instruction set that converts my inputs into in-game actions and also grabs live game data.

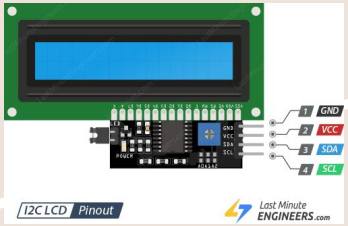


Hardware & Electrical









Primary Components:

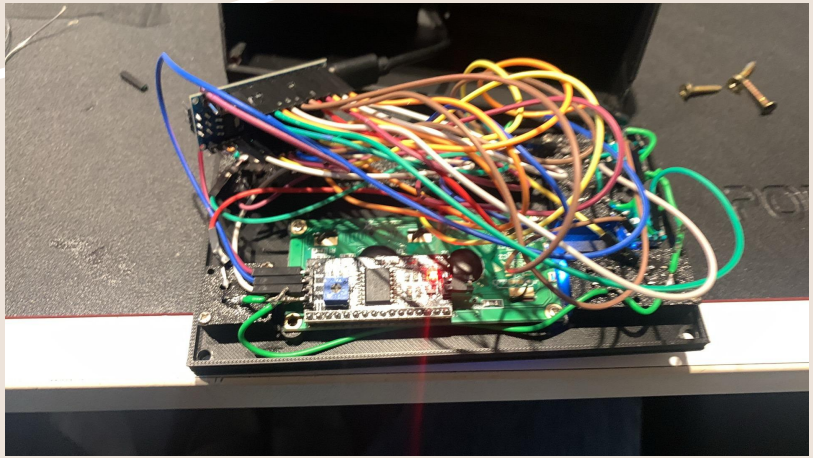
- Arduino Nano
- 6x6 mm buttons x11
- Rotary Encoder w/ Push Button x2
- LCD with I2C backpack
- Jumper Wires

- Daisy chained the Ground wire of each component to save space



Example of daisy-chaining wires

- Biggest Problem:
  - The soldering iron would often burn the 3d printed plastic, so I had to solder each component quickly before it melts the plastic

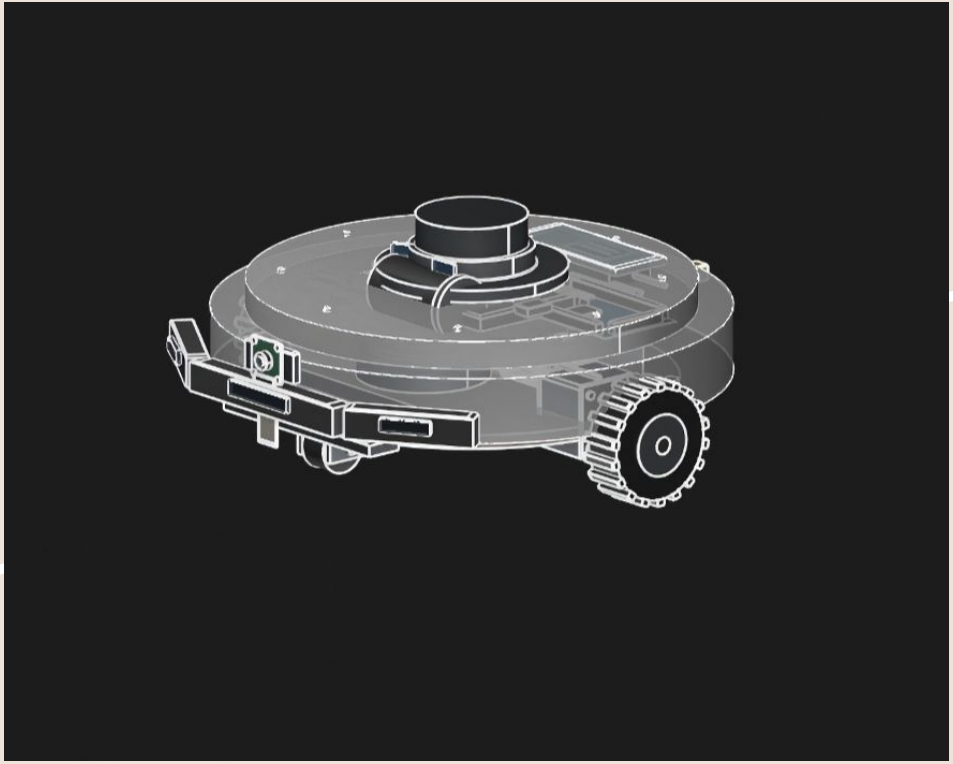


Main Takeaways From this Project

1. My biggest takeaway from this project is the concept of daisy chaining wires. Instead of running individual, redundant ground wires to every single component, daisy chaining made the circuit design more efficient.
    a. **Impact:** Reduced cable clutter, minimized signal clutter, and reduced the risk of short circuits.

2. Overcoming burnt insulation from my FRC soldering experiences taught me proper heat dissipation techniques and soldering techniques.
    a. Learned to use spare connectors/conductive parts as heat sinks
    b. Pre-tin both the wires and metal contacts
    c. Use flux

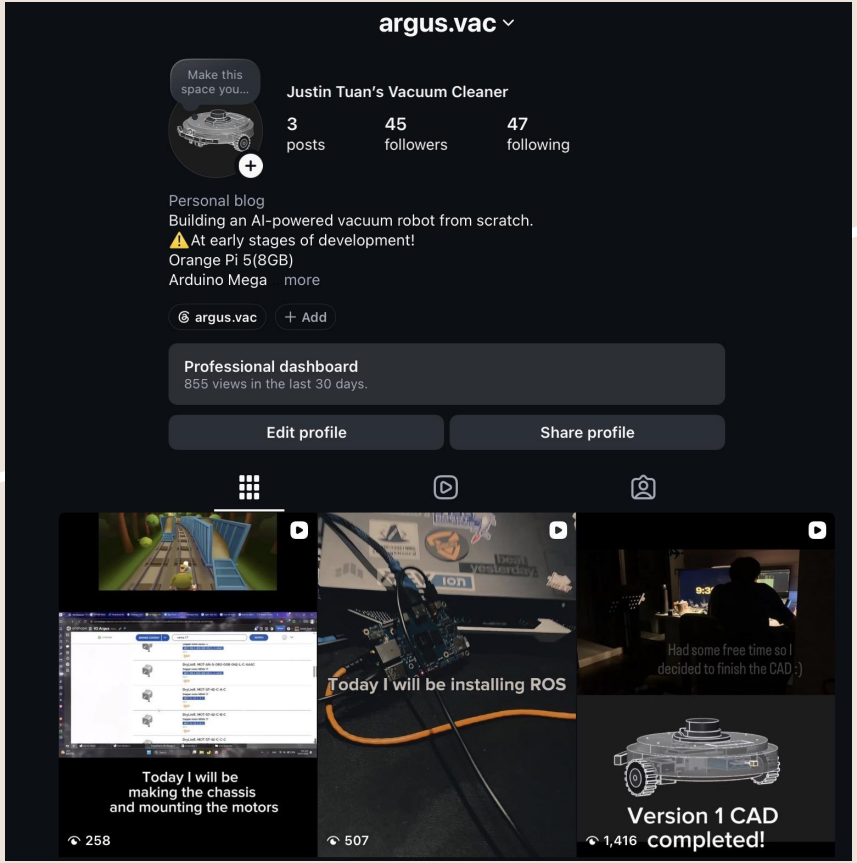
Autonomous Vacuum Cleaner



Instagram Page

This is my Instagram Page that I will be using to document the process of my robot.

@ARGUS.VAC



About

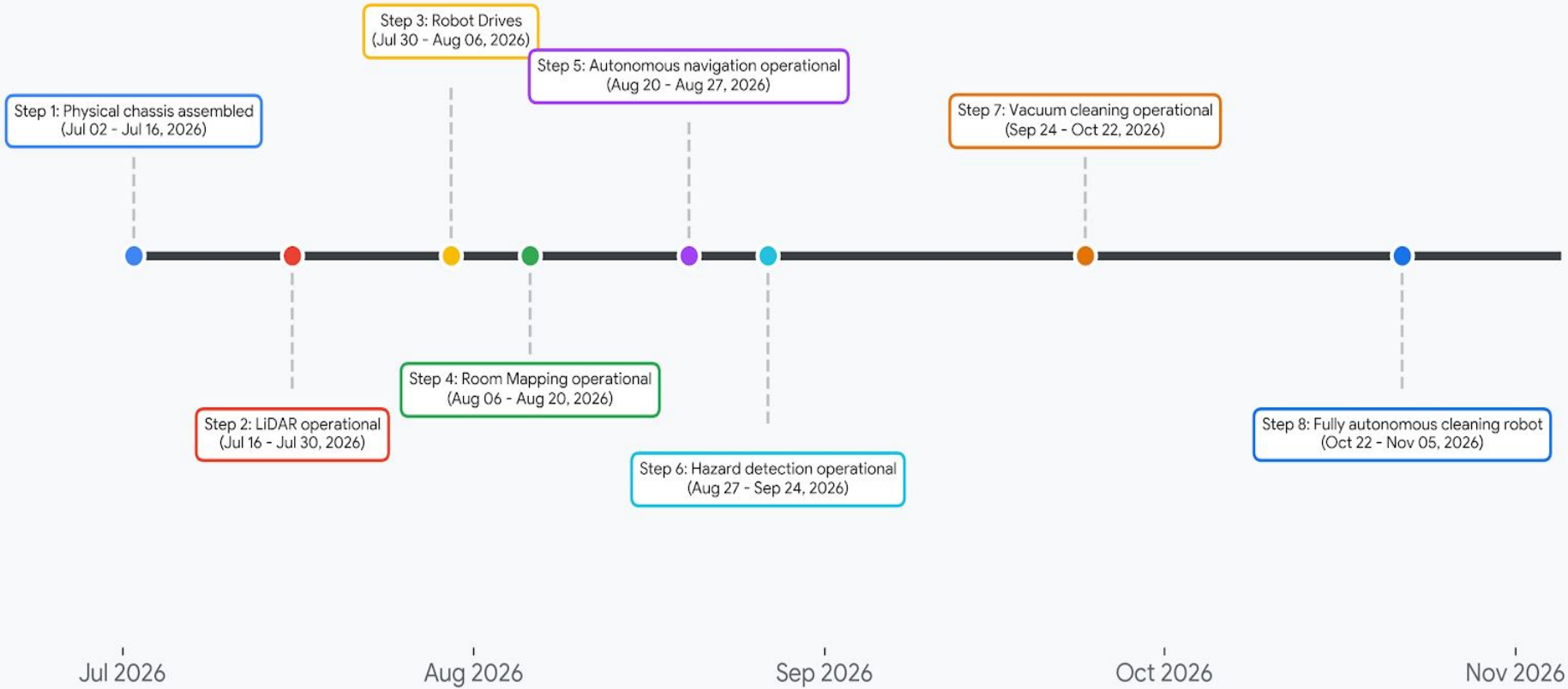
This project will be combining everything I learned in the past few years about electronics, engineering, programming, and communications to

Bill of Materials(Updated)

Recent Fixes

Current Status

Vacuum Robot Project Timeline (2026)



Science Fair